

Supplementary Material

ONCOPLEX: An Oncology-Inspired Hypergraph Model Integrating Diverse Biological Knowledge for Cancer Driver Gene Prediction

Etab Mohammed Alotaibi¹, Omer S Alkhnbashi^{2,3,*}, and Van Dinh Tran^{1,*}

¹Information and Computer Science Department, King Fahd University of Petroleum and Minerals (KFUPM), Dhahran 34462, Saudi Arabia

²Center for Applied and Translational Genomics (CATG), Mohammed Bin Rashid University of Medicine and Health Sciences (MBRU), Dubai Health, Dubai, United Arab Emirates (UAE)

³College of Medicine - Mohammed Bin Rashid University of Medicine and Health Sciences (MBRU), Dubai Health, Dubai, (UAE))

1 Results

1.1 Analysis of newly predicted genes by OncoPlex across 11 cancer types

2 Methods

2.1 Experimental setting

2.1.1 Hyperparameters optimization

A successful machine learning model depends on the selection of suitable hyperparameters, which makes careful tuning essential. We perform hyperparameter optimization using 5-fold cross-validation. The labeled data were divided into K folds, where (K-1) folds were used for training, and one fold was kept for testing. Within each training fold, an inner loop applied a simple grid search to explore combinations of hyperparameters and identify the optimal configuration based on validation performance. We tuned the following hyperparameters:

- Number of hidden units: [64, 128, 256]
- Number of layers: [2, 3, 4]
- Learning rate: [1e-3, 5e-4]
- Weight decay: [1e-3, 1e-4]
- Dropout: [0.25, 0.4, 0.5]
- Positive class weight: [0.2, 0.4, 0.45]

The **outer loop** was then used to evaluate the model with the best hyperparameter combination and report the average and standard deviation for the three metrics AUPRC, AUROC, and the F1 score. This evaluation provides a robust and unbiased estimate of the performance of the model. We repeat the same procedures for the specific cancer setting. In some types of cancer, the best set of hyperparameters is identical, while in other cases, a different set is optimal for that specific cancer.

2.1.2 Datasets

Driver gene labels: We collect driver gene sets for both pan-cancer and cancer-specific driver genes from the following resources:

- Network of Cancer Genes (NCG) v6.0¹: http://ncg.kcl.ac.uk/download_file.php?file=cancergenes_list.txt
- DigSee Database²: <http://210.107.182.61/digseeOld/>
- COSMIC Cancer Gene Census (CGC) v91 and COSMIC Mutation data³: <https://cancer.sanger.ac.uk/cosmic/download>

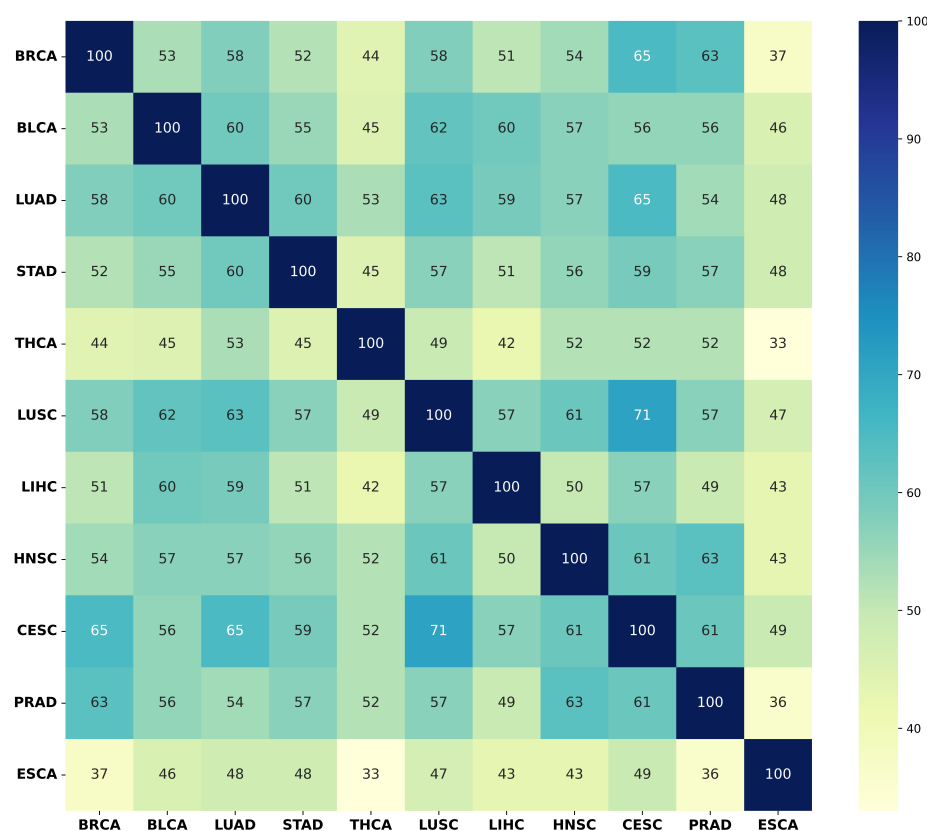


Figure S1. The overlap among the 100 newly predicted genes across different cancer types. For each cancer type, we evaluate how its predicted genes intersect with those of other cancers. As shown, some cancer types, such as BRCA, exhibit high levels of overlap with many others, suggesting shared molecular characteristics. In contrast, cancers like ESCA show minimal overlap with most other types, except specific cases such as CESC, STAD, and LUAD, where moderate gene-level similarities are observed.

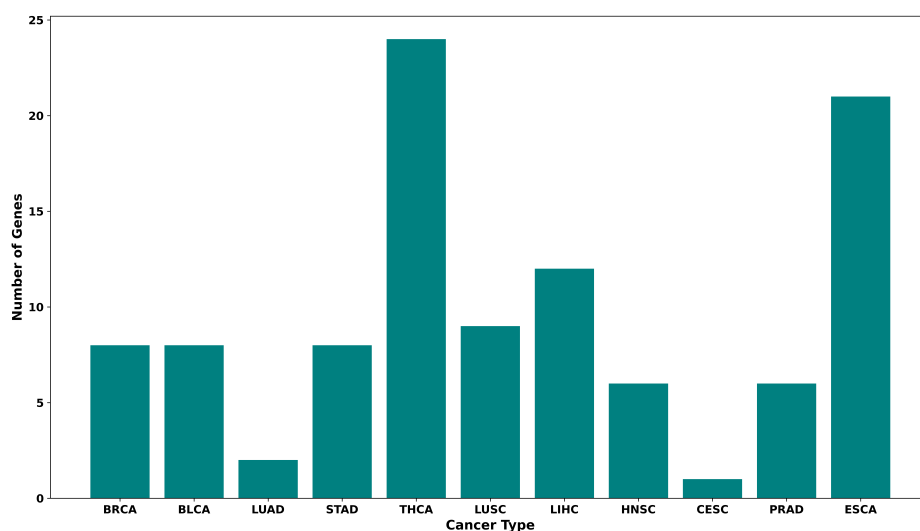


Figure S2. The number of unique predicted genes identified for each cancer type. We analyze the genes that are exclusively predicted for a specific cancer type. Among all cancers, THCA and ESCA exhibit the highest number of unique genes, which can be attributed to their limited overlap in pathways with other cancers. This reduced pathway overlap results in fewer shared genes with other cancer types.

- IntOGen tumor-specific driver lists v2024.09.20⁴ <https://www.intogen.org/download>

Gene sets used for biological validation: To validate our predicted genes, we checked whether our predicted genes appeared in the following cancer gene databases:

- **OncoKB:** A manually curated resource of cancer genes and variants⁵: <https://www.oncokb.org/cancerGenes>
- **ONGene:** A collection of human oncogenes curated from literature⁶.
http://ongene.bioinfo-minzhao.org/ongene_human.txt
- **CancerMine:** A text mining-based database of cancer gene roles extracted from the published literature⁷.

2.2 Node features

We used three types of node features: incidence-based features, core omics features, and comprehensive features.

Incidence-based features: These features were derived from the hypergraph incidence matrix and reflect only the structural connectivity between genes and pathways. Incidence-based features contain no biological information and are used to evaluate how much information comes from the higher-order pathways connection.

Core omics features: These features were derived from three omics modalities: mutation, gene expression, and DNA methylation. Following⁸, the preprocessing steps are described below.

- **Single nucleotide variants:** We extracted the SNV profiles from TCGA MAF files. We used non-silent somatic mutations and removed hypermutated samples to reduce noise due to genome instability.
- **RNA-Seq gene expression data:** TCGA provided RNA-seq expression profiles as FPKM counts. After batch correction, we computed the logarithmic

3 Supplementary Tables

The following supplementary tables (Supplementary Tables S1–S12) are provided as a single Excel file located in the same folder as this document, with each table presented in a separate worksheet.

References

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